



Swapped the Card, Not the Habit

Session length and peak occupancy in the gym sauna before and after a safety-scare timer swap

Joost Nwosu · Sten Okwuosa — School of Continuous Improvement

The problem

Campus Recreation replaced the sauna's honesty-based flip-timer card with a swipe-logged app timer after a member reported losing track of time inside. Does logging entry and exit electronically change how long anyone stays, or just how confidently the duration gets reported?

Research questions

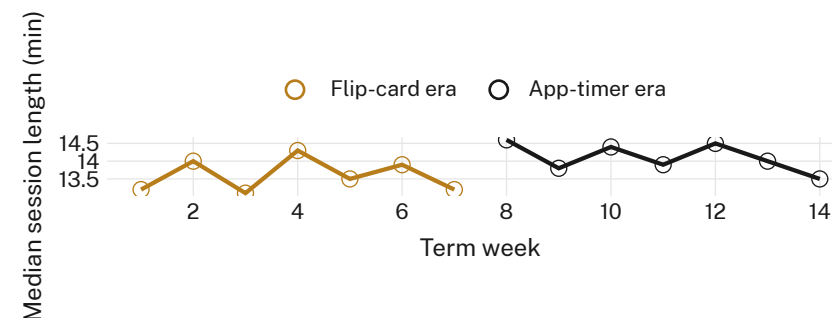
- Does median session length change after the app-timer rollout?
- Does peak concurrent occupancy change across the same two periods?
- Do flip-card and app-logged durations agree during a shared overlap?

Approach

- $n = 2,214$ sauna entries, 14 weeks (7 before / 7 after rollout); Adaptive Metrics Lab occupancy sensor, 1-min sampling
- Session length: swipe interval; flip-card era from the honesty board's self-logged sign-out time
- Posted 15-min limit and towel-hook count held constant throughout

Findings

Median session length holds at 14 minutes across both eras (13.6 min before, 14.1 after; Mann-Whitney $p = 0.41$). Peak occupancy is unchanged at six.



Weekly median sauna session length holds within 13.1–14.6 minutes across all 14 weeks of Semester 1 2026, showing no shift at the Week 8 swipe-timer changeover.

Discussion & conclusions

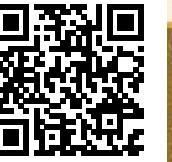
The swap changed who could see the record, not how long anyone stayed. Flip-card and swipe-log durations agree closely over a four-week overlap ($r = 0.71$). **Next:** whether the log changes how Facilities schedules towel replenishment, which the flip-card count never fed.

Selected references

1. Li, N., Calis, G., & Becerik-Gerber, B. (2012). Measuring and monitoring occupancy with an RFID based system for demand-driven HVAC operations. *Automation in Construction*, 24, 89–99. doi:10.1016/j.autcon.2012.02.013
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3. Waddell, K. J., & Lang, C. E. (2018). Comparison of self-report versus sensor-based methods for measuring the amount of upper limb activity outside the clinic. *Archives of Physical Medicine and Rehabilitation*, 99(9), 1913–1916. doi:10.1016/j.apmr.2017.12.025
4. Laukkanen, T., et al. (2018). Sauna bathing is associated with reduced cardiovascular mortality and improves risk prediction in men and women: a prospective cohort study. *BMC Medicine*, 16, 219. doi:10.1186/s12916-018-1198-0

De-identified entry/exit timestamps only; no member names or membership numbers retained. Sauna occupancy cap unchanged throughout (8 persons).

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“Fourteen minutes, logged or not.”

Median sauna session length is unchanged across the swipe-timer rollout, 14 weeks, Semester 1 2026.